

EARSHOT — everything needed to write the abstract

- **For:** Adrian Erlikhman — **final numbers**, Saturday 12 September 2026
- **Venue:** AI for Peace workshop @ NeurIPS 2026, Paris, 12–13 December
- **Deadline:** Mon 21 Sept AoE = **Tue 22 Sept, 05:00 Pacific**
- **Repo:** <https://github.com/adrian-erlikhman/earshot> (public)

1. The container you are writing into

Submission	Google Form. Pasted text. No file upload.
Abstract field	2,500 characters max. References count against it.
TL;DR field	300 characters max , separate
Figures	Cannot be submitted. Poster only.
Review	Single-blind
Attendance	Gating on the form; no travel funding. Adrian presents.

2. Where the evidence landed — read this first

Surveillance-classified patents are **rare** among patents citing NLP research, and once you compare like with like — patents granted in the same era — they are **not distinguishable** from patents citing science in general. No NLP subfield stands out, and there is no time trend. The earlier treatment-versus-control gap was produced by grant era, not by NLP.

So the paper is about what the patent record can and cannot show — the framing we fixed before any results existed — plus three method findings that anyone doing this kind of analysis needs.

Every number in this brief comes from a random sample and has been checked against the results files by `tools/verify_brief.py`. Numbers in earlier briefs came from a sample truncated by grant date and are withdrawn (section 7).

3. Pipeline numbers — locked

#	Number	Meaning
1	127,851	ACL Anthology papers, 1952–2026
2	69,327	resolve to OpenAlex
3	26,085	patent-to-paper citation links
4	9,048	distinct USPTO records citing an ACL paper (8,553 granted patents, 487 pre-grant application publications, 8 other)
5	24,829 vs 24,821	our replication of Zhang's core-venue paper count — eight apart

#	Numb er	Meaning
6	0.33%	ACL-citing records as a share of the 2,748,170 science-citing records in Reliance on Science v65
7	91.6%	of our citation links are front-page only (62.3% across all patents)

Correction. Earlier material said v65 covers only 2015–2025 grants. That is wrong: it includes patents granted before 1976 and pre-grant application publications.

4. Results — random samples, assignee hidden

Treatment: 2,500 drawn at random from the 9,048; **2,374 classified**. The other 126 are pre-grant application publications (125), which the text source does not serve, plus 1 fetch failure. **Control:** 2,500 drawn uniformly from records citing science but no ACL paper; 2,241 classified.

Measure	Value
Surveillance, NLP-citing patents	11 of 2,374 = 0.46% (95% CI 0.25–0.80%)
Dual-use / ambiguous	3 = 0.13%
Military / defence	1 = 0.04%
Surveillance, science-citing control (crude)	6 of 2,241 = 0.27%
Control reweighted to the treatment's grant-era mix	0.41%
Era-matched odds ratio (Mantel–Haenszel)	1.03 (95% CI 0.38–2.78), p = 0.95
NLP subfields significant for surveillance	0 of 10 (Benjamini–Hochberg q < 0.05)
By grant year	0.00% (2015–17), 0.83% (2018–20), 0.66% (2021–23), 0.47% (2024–26) — no trend

Why era matching is necessary

Grant era	NLP-citing	Science-citing
1976–2014	0 of 417	2 of 1,110
2015 – May 2021	3 of 634 (0.47%)	2 of 616 (0.32%)
May 2021 onward	8 of 1,322 (0.61%)	2 of 386 (0.52%)

Surveillance-classified patents are recent. NLP-citing patents are recent: 58% were granted May 2021 or later, against 26% of science-citing patents. A crude comparison mostly measures that difference. **Never quote the crude ratio.**

The interval 0.38–2.78 is the honest bound: we cannot rule out a rate up to about 2.8 times higher, or well below. Say "not distinguishable", not "the same".

5. The three method findings — the heart of the abstract

A. Applicant names sway LLM labels, in both directions. We re-classified 99 patents with the assignee shown and hidden (every non-neutral label plus every defence-sounding applicant). **10 of 31 non-neutral labels (32%) changed.** Defence-sounding applicants pulled labels toward military: 6 of the 7 labels that dropped to neutral when the name was hidden had been military (a botulism antitoxin from a military medical academy was the clearest). In the other

direction, 9 patents called neutral with the applicant visible were flagged once it was hidden, among them patents from large consumer-technology and healthcare firms. The model is deterministic: the redacted re-call reproduced 99 of 99 labels. Hiding the assignee fixes this by construction.

B. Technology classes are a poor proxy for surveillance. Of **69** patents that examiners placed in surveillance-related technology classes (speaker recognition, biometrics, alarms, access control, CCTV), the rubric codes **2 as surveillance**, 3 as dual-use and **64 as neither**. Nearly half of those 64 are voice or digital assistants. The main class, G10L17, is speaker recognition, and one of its subgroups covers interactive voice interfaces — so class-based or keyword-based counting, the approach used in the computer-vision work, would sweep these in. Examiners never judged these patents to be surveillance; the overcount is in using their classes as a proxy. **Not yet human-validated — the gold set tests exactly these boundary cases.**

C. The record is thin where it is most cited. Under half of one percent, no subfield concentration, no trend, and no detectable difference from science-citing patents of the same era. The evidence the dual-use debate leans on cannot carry the weight put on it.

One concrete example is worth a clause if space allows: **HRL Laboratories' patent for early detection of civil unrest from social media** is classified surveillance with the assignee hidden.

6. The limits — put them early

- **91.6% of links are front-page citations**, which show prior-art disclosure, not use. We measure proximity, not transmission.
- **Patent abstracts often omit the application**, so a low rate bounds what the record can show, not what the field does.
- **Granted patents only.** Pre-grant application publications (about 5% of NLP-citing records) could not be retrieved.
- **Power.** With 11 events the comparison is a bound. The interval is the result.

7. Withdrawn — do not reuse any of these

Withdrawn	Why
0.69%, 1.02%, 0.81% surveillance	truncated-by-grant-date samples (fetched in sorted id order)
control 0.27% and risk ratio 1.95 or 2.11	crude, era-confounded
"25% of labels flip, over-labelling"	computed on the truncated set; now 32%, both directions
"21 of 26 examiner-class patents"	truncated set; now 64 of 69
"2015–2025 grants"	false description of the data
positive control 6 of 7; redaction 0 of 60	circular; uninformative
in-text ablation as support	0 of 292, upper bound 0.86%, above the overall rate

8. What the method paragraph must contain

- **Operator vs subject** and **authentication vs identification**.
- **Assignee hidden** from the model and both human coders.
- **Two coders, the authors**; agreement is internal consistency, not independent validation.
- Per-class F1 and Gwet's AC1 alongside kappa, since one class dominates.

9. Sunday 13 September — labelling session

- 200 patents, title and abstract only. **Do not look patents up. Do not open** `results/patent_labels.csv` — it is public and holds the model's labels.
- The sheet was built before the sampling fix, so its items skew toward 2018–2023 grants. That is fine for measuring agreement; it is not used for population rates.
- **Your coding is not blind on the surveillance items** — you have seen the model's labels for several. Michael's labels are the reference for those.

10. Titles

- **Proximity, Not Transmission: What the Patent Record Can and Cannot Show About NLP and Surveillance** — recommended
- Authentication Is Not Identification: Classifying the Patents That Cite Language-Technology Research
- Earshot: What Are the Patents Citing NLP Research Actually For?

11. TL;DR (checked against the 300-character limit)

Earshot links 127,851 ACL papers to the 9,048 USPTO patents citing them and classifies what those patents do. Surveillance is rare (0.46%) and not detectably more common than in science-citing patents of the same era; applicant names sway LLM labels.

12. Draft abstract (checked against the 2,500-character limit)

Only the agreement numbers are pending (Sunday).

Kalluri et al. (2025) traced computer-vision research into surveillance patents. Whether language technology follows the same path is often asserted but rarely measured, and patent citations are easy to overread. We ask what the public patent record can reliably show. Earshot links 127,851 ACL Anthology papers through OpenAlex to the 9,048 USPTO patents that cite them, matching Zhang's (2025) paper count for the core venues (24,829 vs. 24,821), and classifies what those patents do. 91.6% of these citations appear only on a patent's front page. Our rubric separates a system's operator from its subject, and authentication from identification: verifying a claimed identity is not surveillance, while picking a person out of a population can be. Assignee names are hidden from the model and from two human coders (the authors); revealing them changed 32% of the model's non-neutral labels, in both directions. Of 69 patents examiners placed in surveillance-related technology classes, the rubric codes 2 as surveillance; nearly half of the rest are voice assistants. Coders agree with the model at [surveillance $F1 = x$; $AC1 = y$] on a stratified sample. In a random sample of 2,374 granted citing patents, 0.46% (95% CI 0.25–0.80%) are classified as surveillance, no subfield stands out, and the rate is not distinguishable from science-citing patents of the same grant era (odds ratio 1.03, 95% CI 0.38–2.78). These results do not establish whether NLP flows into surveillance: patent abstracts often omit application detail, and citations show proximity, not transmission. Earshot shows how dual-use pathways can be studied without mistaking technical similarity, applicant identity, or citation for evidence of deployment.

13. Citations (verified — see refs.bib)

- **Kalluri et al., Nature 2025** — computer vision to surveillance patents. Cite the Nature title. Hand-coded 100 papers and 100 patents, then used a keyword list.
- **Zhang, ACL 2025** — linked ACL papers to patents first. Never claim to be first.

- **Alcácer, Gittelman and Sampat, Research Policy 2009** — examiner citations.
- **Marx and Fuegi** — Reliance on Science.
- **Srivastava et al., AI for Peace at ICLR 2026** — in this workshop's own first edition.
- **Garcia and Katirai, FAcCT 2026** — the computer-vision version. Noa Garcia chairs this workshop.

Spend so far: under \$1.50 of the \$50 budget.